Supplementary Information

Integrative structural bioinformatics and molecular dynamics analysis of water-soluble synaptic vesicle protein QTY-variants reveals synaptic vesicle constraints and evolutionary coupling of T<=>V

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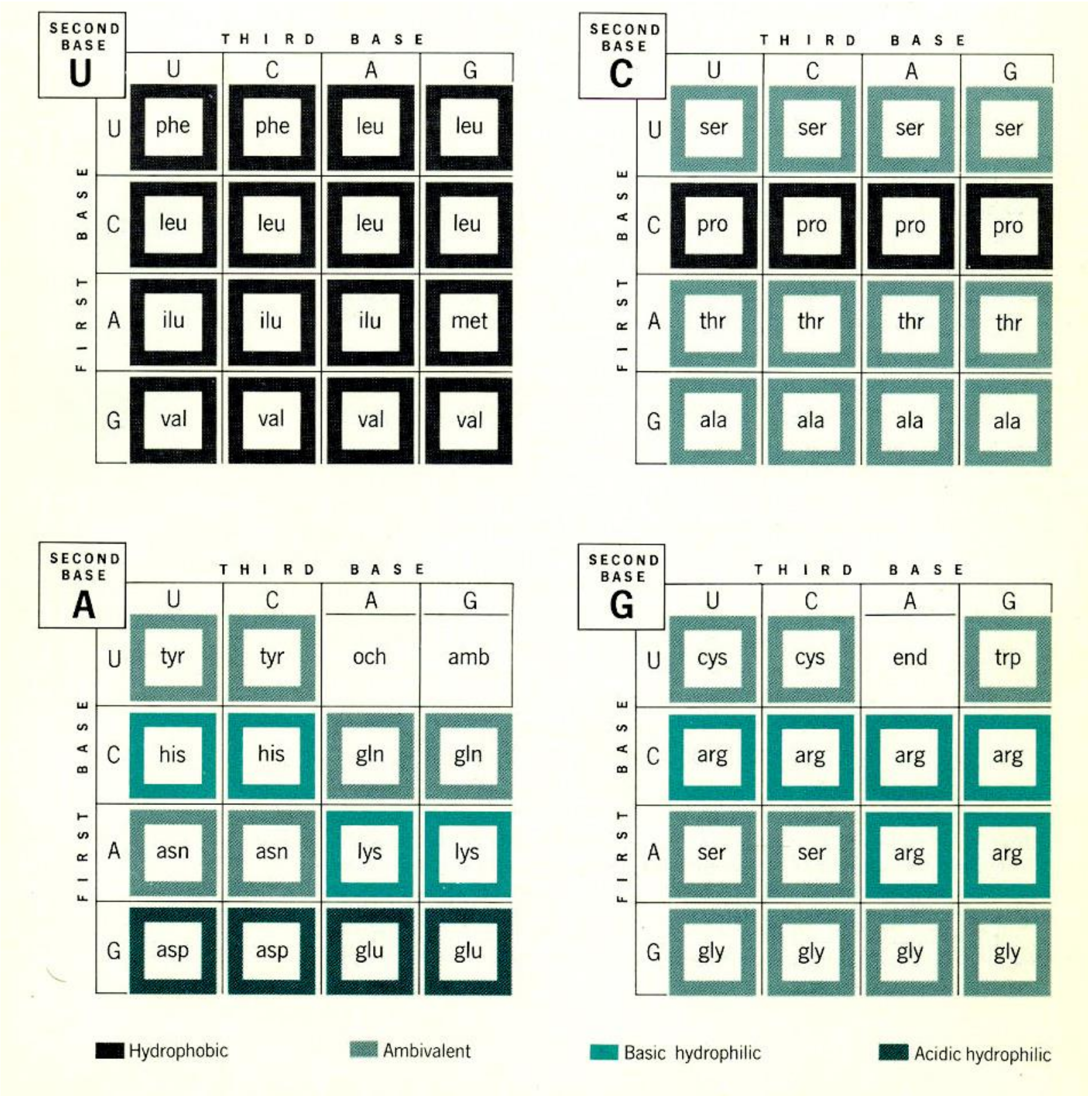


Figure S1. The chemical nature of amino acids is primarily determined by the second position of their genetic code. The second position is particularly significant, as seen in several examples:

- **i.** Amino acids with uracil (U) in the second position tend to be hydrophobic, including phenylalanine (Phe), leucine (Leu), isoleucine (Ile), valine (Val), and methionine (Met).
- **ii.** Amino acids with cytosine (C) in the second position are either less hydrophobic, such as proline (Pro) and alanine (Ala), or contain a hydroxyl (-OH) group, like serine (Ser) and threonine (Thr).
- **iii.** Amino acids with adenine (A) in the second position are generally hydrophilic and water-soluble. These include aspartic acid (Asp), glutamic acid (Glu), asparagine (Asn), lysine (Lys), histidine (His), and tyrosine (Tyr). Additionally, two stop codons, ochre (UAA) and amber (UAG), have A in the second position.
- **iv.** Amino acids with guanine (G) in the second position are also water-soluble. Examples are arginine (Arg) and serine (Ser). Cysteine (Cys) is partially water-soluble, and glycine (Gly) is achiral with a hydrogen atom as its side chain. The stop codon UGA is also noted here.

Overall, the presence of pyrimidines (U and C) at the second position is associated with hydrophobic properties, whereas purines (A and G) contribute to hydrophilicity.

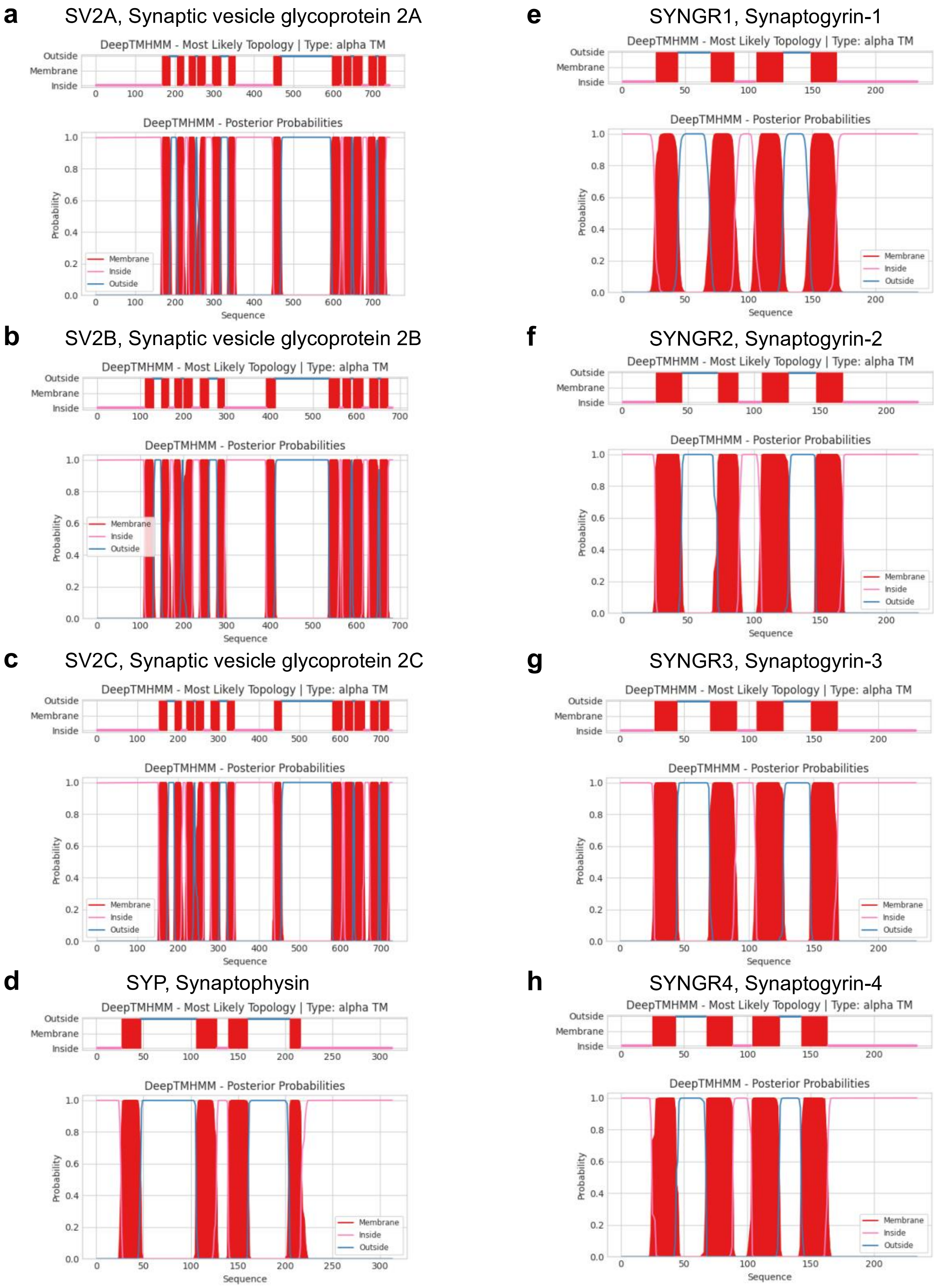


Figure S2. Membrane topology models of eight native synaptic vesicle proteins. Transmembrane (TM) helix predictions were carried out using DeepTMHMM, a deep learning model for transmembrane topology prediction and classification.

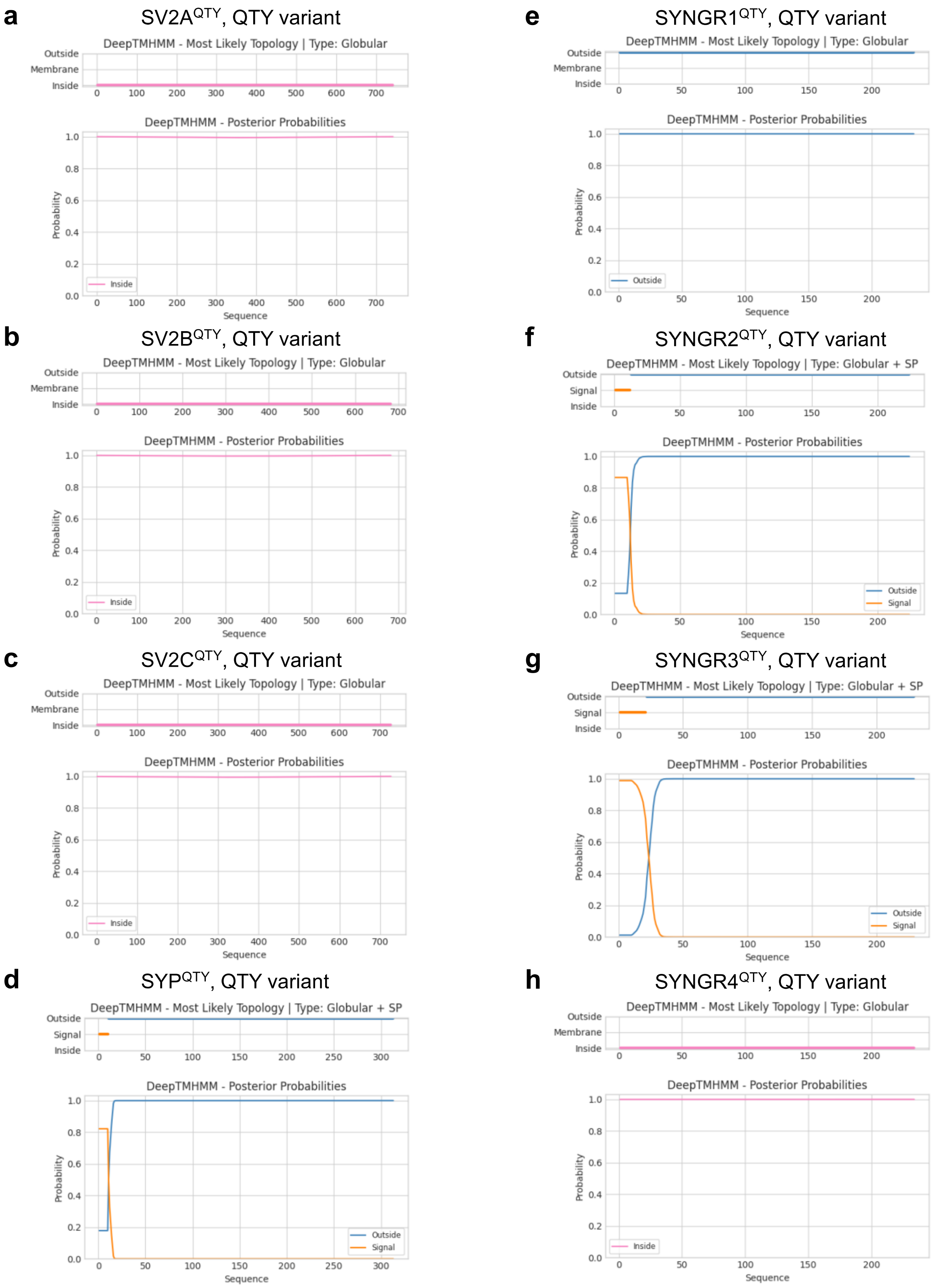


Figure S3. Membrane topology models of QTY variants of eight synaptic vesicle proteins. Transmembrane (TM) helix predictions were carried out using DeepTMHMM, a deep learning model for transmembrane topology prediction and classification.

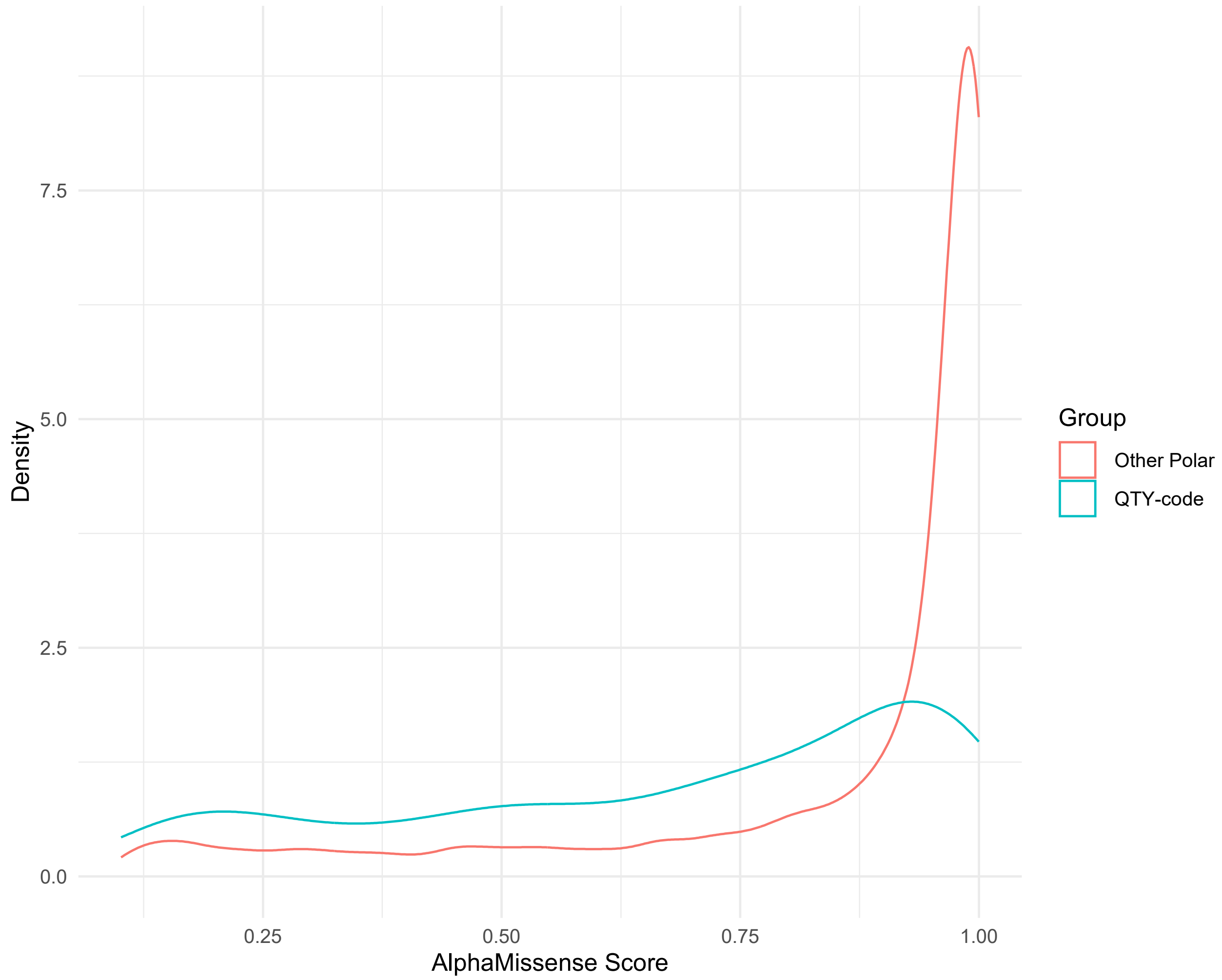


Figure S4. Density Plot of predicted effects of variations at the residues L, I, F of Synaptic vesicle glycoproteins A-C (SV2A-C). The x-axis indicates the scores of the AlphaMissense predicted effect of the substitution, ranging from benign (0.0) to damaging (1.0). The y-axis indicates the density of the score. Median score for QTY variants was M=0.752 (0.470-0.926), median score for other polar substitutions was M=0.972 (814-0.994). Kolmogorov Smirnov test indicated a significant difference between the two distributions (D(477,4288) = 0.4, p =<0.001).

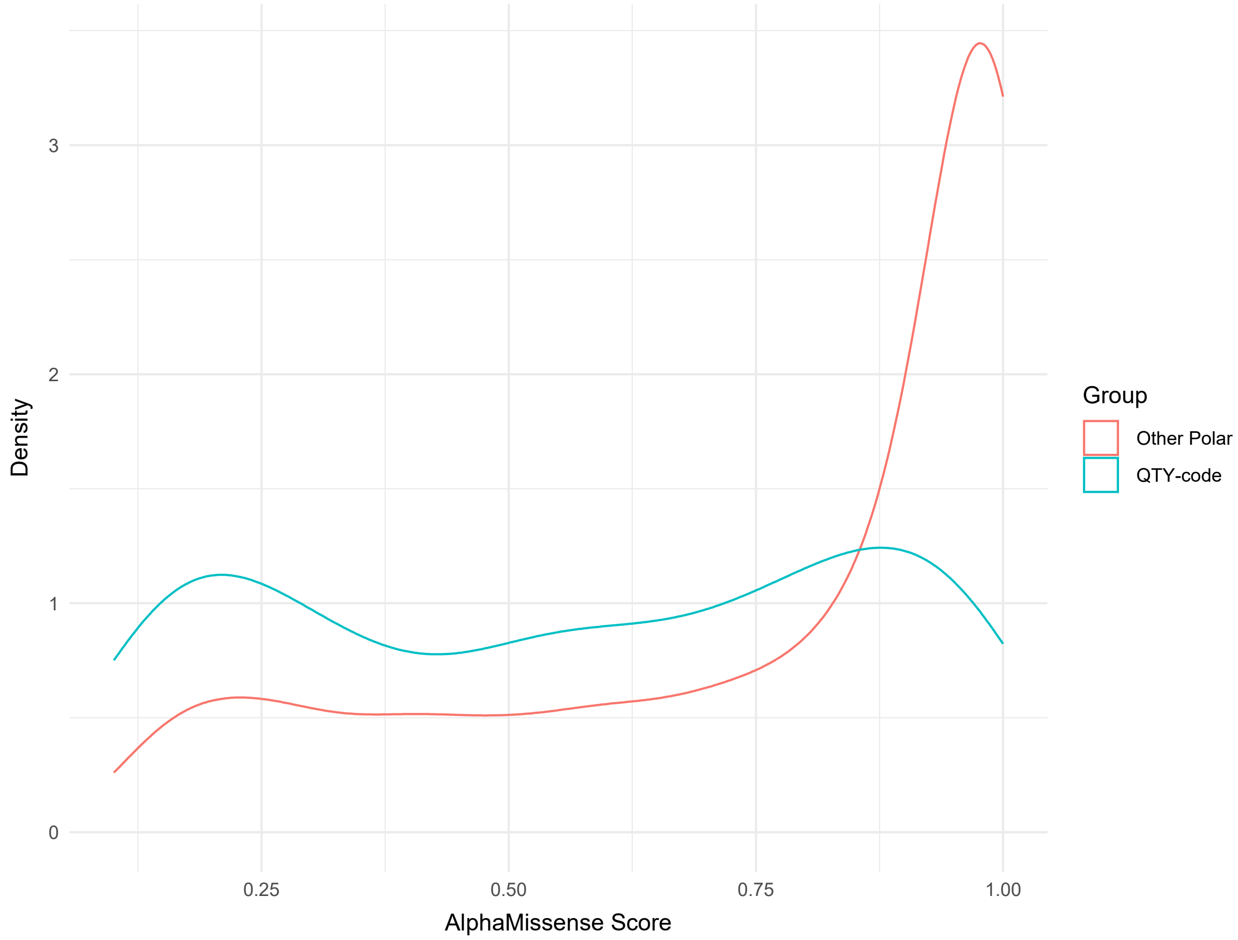


Figure S5. Density Plot of predicted effects of variations at the residues L, I, F of Synaptogyrins (SYNGR1-4) and Synaptophysin (SYP). The x-axis indicates the scores of the AlphaMissense predicted effect of the substitution, ranging from benign (0.0) to damaging (1.0). The y-axis indicates the density of the score. Median score for QTY variants was M= 0,580 (0,286-0,833), median score for other polar substitutions was M=0,899 (0,570-0,989). Kolmogorov Smirnov test indicated a significant difference between the two distributions (D(248,2229) = 0.34, p <0.001).

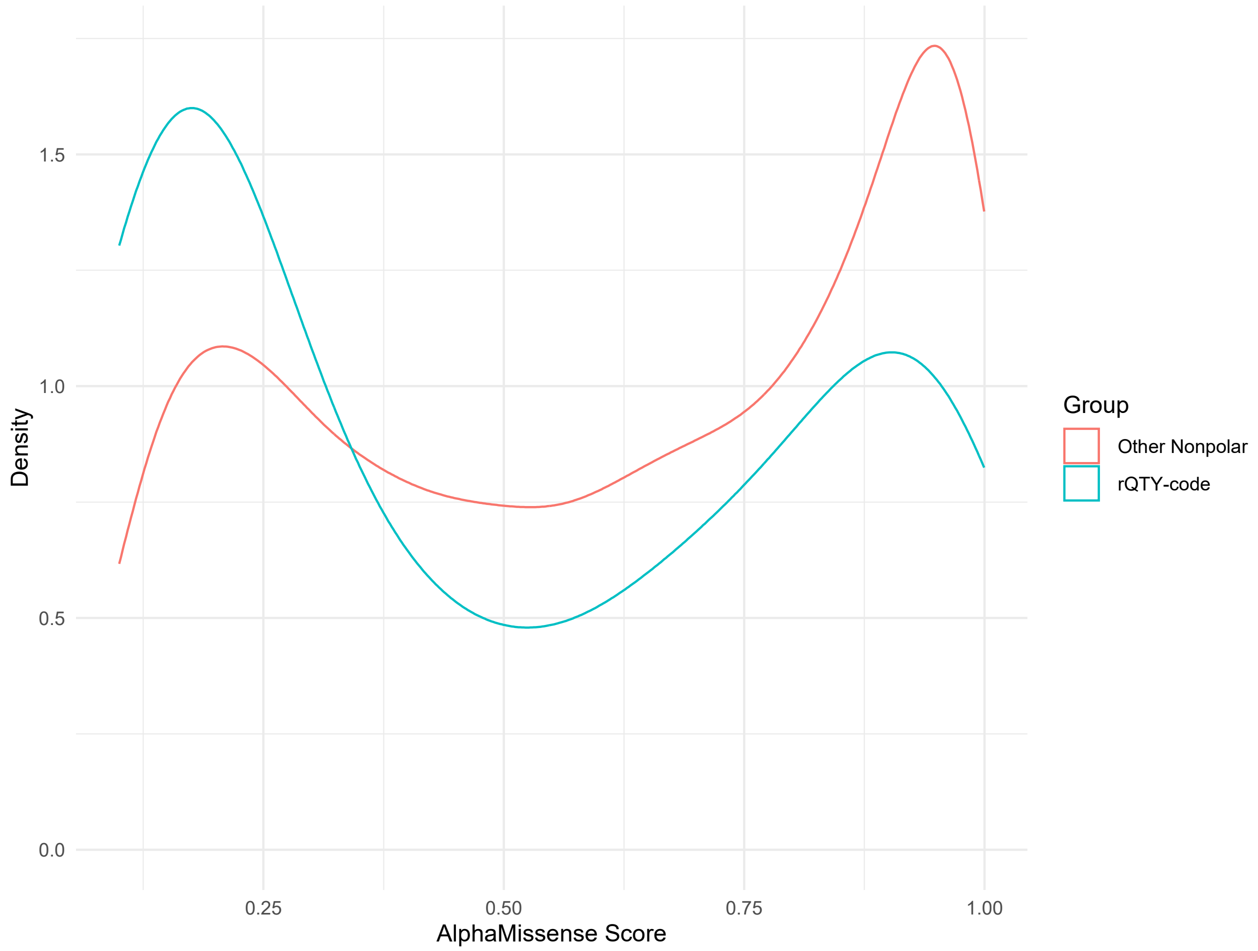


Figure S6. Density Plot of predicted effects of variations at the residues Q, T, Y of Synaptic vesicle glycoproteins A-C (SV2A-C). The x-axis indicates the scores of the AlphaMissense predicted effect of the substitution, ranging from benign (0.0) to damaging (1.0). The y-axis indicates the density of the score. Median score for rQTY variants was M=0.399 (0.175-0.842) median score for other nonpolar substitutions was M=0.652 (0.324-0.895). Kolmogorov Smirnov test indicated a significant difference between the two distributions (D(2337,260) = 0.22, p <0.001).

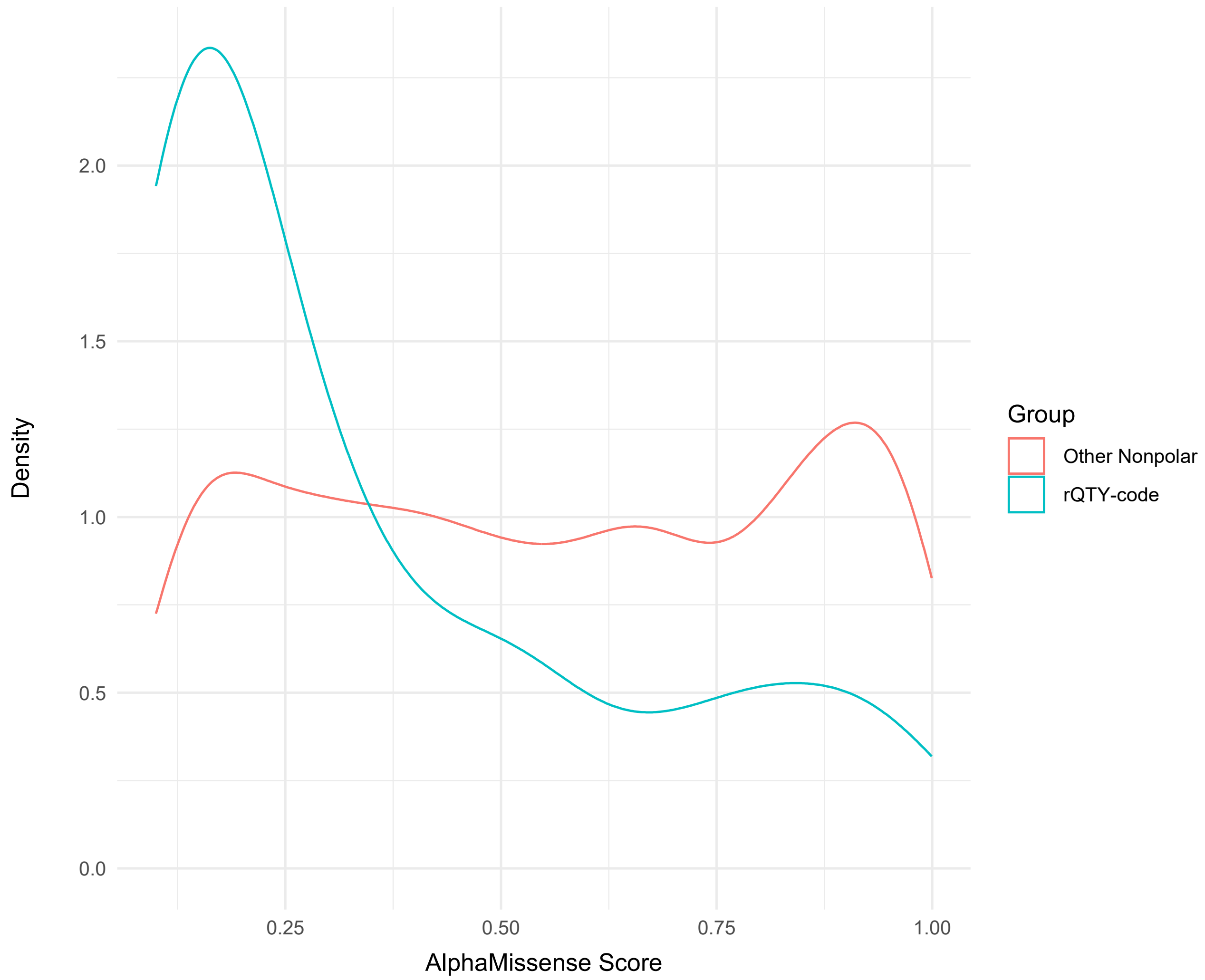


Figure S7. Density Plot of predicted effects of variations at the residues Q, T, Y of Synaptogyrins (SYNGR1-4) and Synaptophysin (SYP). The x-axis indicates the scores of the AlphaMissense predicted effect of the substitution, ranging from benign (0.0) to damaging (1.0). The y-axis indicates the density of the score. Median score for rQTY variants was M= 0.240 (0.141-0.519) median score for other nonpolar substitutions was M= 0.8993 (0.570-0.989). Kolmogorov Smirnov test indicated a significant difference between the two distributions (D(1655,184) = 0.35, p <0.001)

Figure S8. Synaptic vesicle glycoprotein 2A (SV2A) residue-wise evolutionary conservation profiles. Evolutionary conservation grades of each amino acid residue predicted by ConSurf server; visualized by the color-coding scheme of nine colors, ranging from turquoise (variable) through white (average) through burgundy (conserved) represents conservation grades 1 to 9, in order of increasing conservation (1= Variable, 5= Average, 9= Conserved). Conservation grades were calculated for the source amino acid sequence and the corresponding Alphafold3 predicted native structure. For clarity, the N- and C-termini and large loops were deleted.

Figure S9. Synaptic vesicle glycoprotein 2B (SV2B) residue-wise evolutionary conservation profiles. Evolutionary conservation grades of each amino acid residue predicted by ConSurf server; visualized by the color-coding scheme of nine colors, ranging from turquoise (variable) through white (average) through burgundy (conserved) represents conservation grades 1 to 9, in order of increasing conservation (1= Variable, 5= Average, 9= Conserved). Conservation grades were calculated for the source amino acid sequence and the corresponding Alphafold3 predicted native structure. For clarity, the N- and C-termini and large loops were deleted.

Figure S10. Synaptic vesicle glycoprotein 2C (SV2C) residue-wise evolutionary conservation profiles. Evolutionary conservation grades of each amino acid residue predicted by ConSurf server; visualized by the color-coding scheme of nine colors, ranging from turquoise (variable) through white (average) through burgundy (conserved) represents conservation grades 1 to 9, in order of increasing conservation (1= Variable, 5= Average, 9= Conserved). Conservation grades were calculated for the source amino acid sequence and the corresponding Alphafold3 predicted native structure. For clarity, the N- and C-termini and loops were deleted.

Figure S11. Synaptophysin (SYP) residue-wise evolutionary conservation profiles. Evolutionary conservation grades of each amino acid residue predicted by ConSurf server; visualized by the color-coding scheme of nine colors, ranging from turquoise (variable) through white (average) through burgundy (conserved) represents conservation grades 1 to 9, in order of increasing conservation (1= Variable, 5= Average, 9= Conserved). Conservation grades were calculated for the source amino acid sequence and the corresponding Alphafold3 predicted native structure. For clarity, the N- and C-termini and large loops were deleted.

Figure S12. Synaptogyrin-1 (SYNGR1) residue-wise evolutionary conservation profiles. Evolutionary conservation grades of each amino acid residue predicted by ConSurf server; visualized by the color-coding scheme of nine colors, ranging from turquoise (variable) through white (average) through burgundy (conserved) represents conservation grades 1 to 9, in order of increasing conservation (1= Variable, 5= Average, 9= Conserved). Conservation grades were calculated for the source amino acid sequence and the corresponding Alphafold3 predicted native structure. For clarity, the N- and C-termini and large loops were deleted.

Figure S13. Synaptogyrin-2 (SYNGR2) residue-wise evolutionary conservation profiles. Evolutionary conservation grades of each amino acid residue predicted by ConSurf server; visualized by the color-coding scheme of nine colors, ranging from turquoise (variable) through white (average) through burgundy (conserved) represents conservation grades 1 to 9, in order of increasing conservation (1= Variable, 5= Average, 9= Conserved). Conservation grades were calculated for the source amino acid sequence and the corresponding Alphafold3 predicted native structure. For clarity, the N- and C-termini and large loops were deleted.

Figure S14. Synaptogyrin-3 (SYNGR3) residue-wise evolutionary conservation profiles. Evolutionary conservation grades of each amino acid residue predicted by ConSurf server; visualized by the color-coding scheme of nine colors, ranging from turquoise (variable) through white (average) through burgundy (conserved) represents conservation grades 1 to 9, in order of increasing conservation (1= Variable, 5= Average, 9= Conserved). Conservation grades were calculated for the source amino acid sequence and the corresponding Alphafold3 predicted native structure. For clarity, the N- and C-termini and large loops were deleted.

Figure S15. Synaptogyrin-4 (SYNGR4) residue-wise evolutionary conservation profiles. Evolutionary conservation grades of each amino acid residue predicted by ConSurf server; visualized by the color-coding scheme of nine colors, ranging from turquoise (variable) through white (average) through burgundy (conserved) represents conservation grades 1 to 9, in order of increasing conservation (1= Variable, 5= Average, 9= Conserved). Conservation grades were calculated for the source amino acid sequence and the corresponding Alphafold3 predicted native structure. For clarity, the N- and C-termini and large loops were deleted.

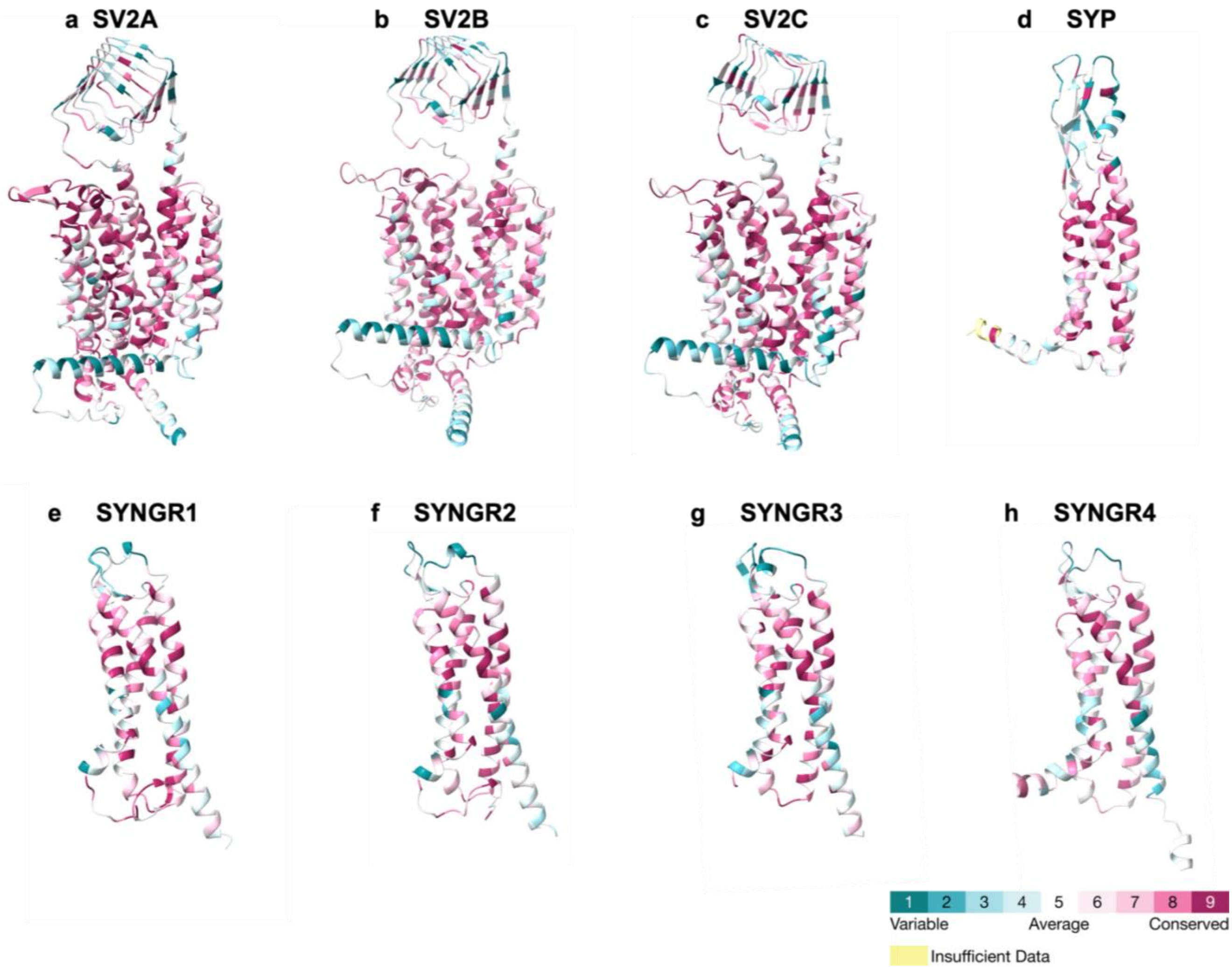
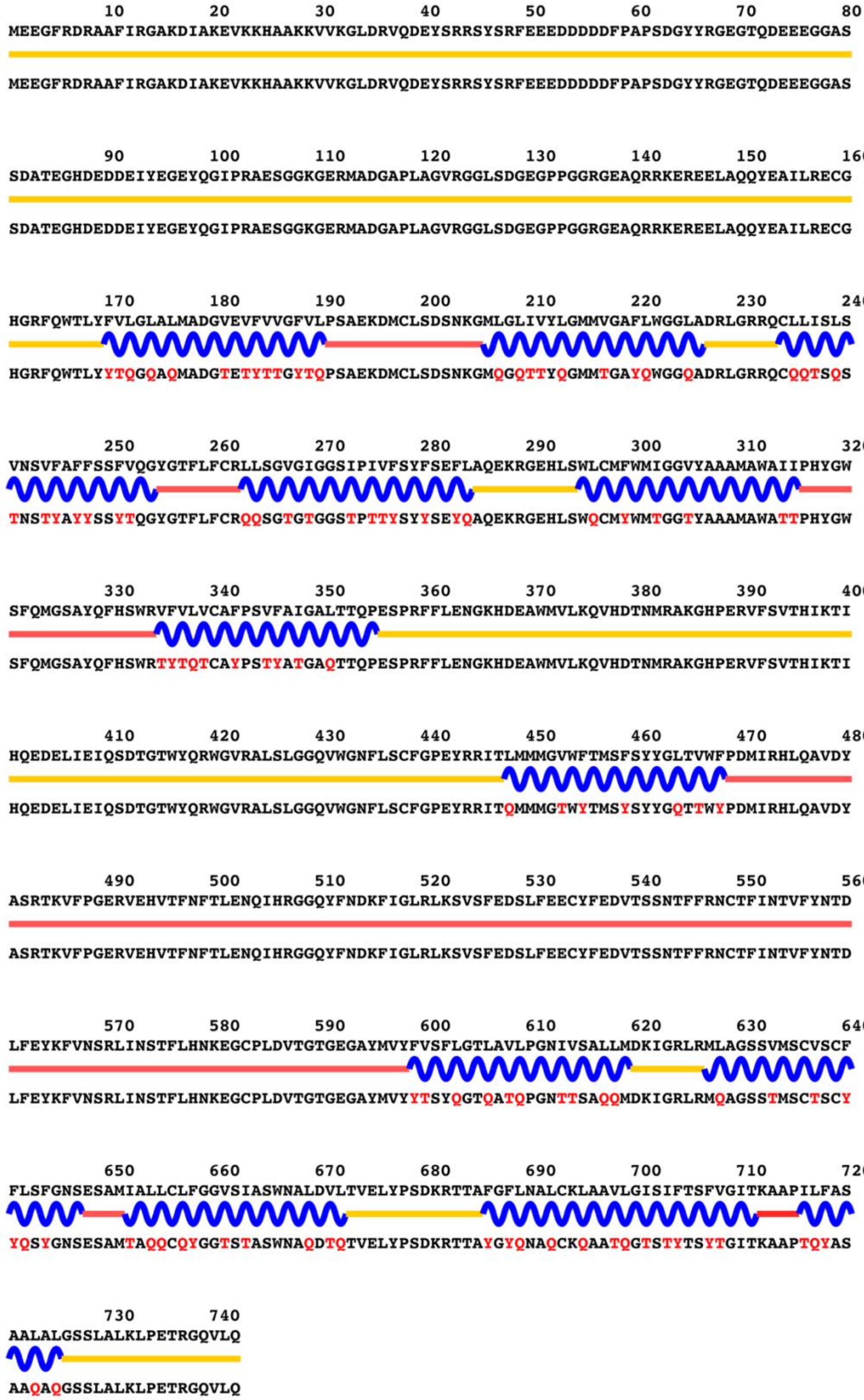
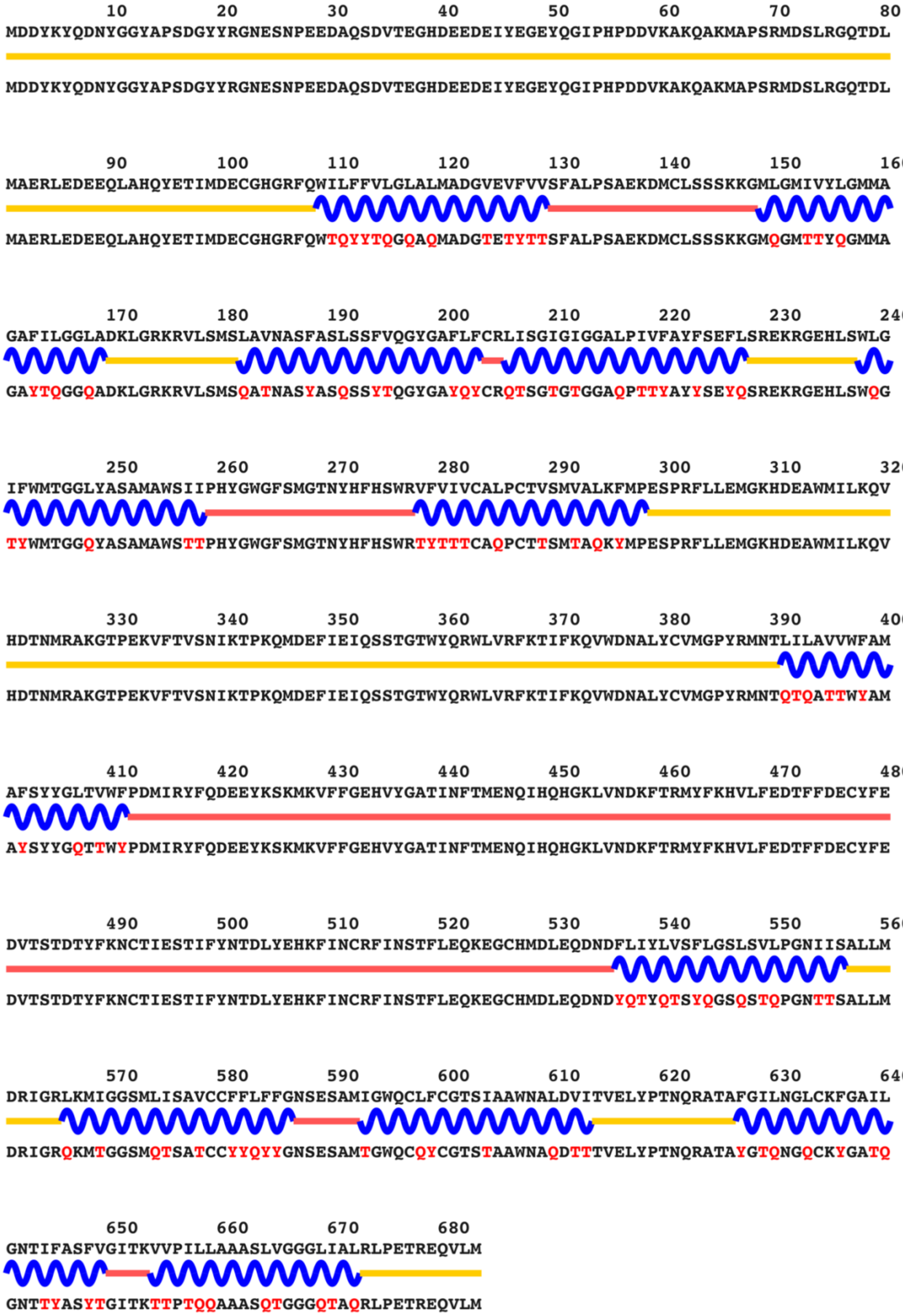


Figure S16. **Evolutionary conservation profiles of 8 synaptic vesicle proteins including synaptic vesicle glycoprotein 2 family, synaptophysin, synaptogyrins.** The predicted native structures and their residues colored by evolutionary conservation grades. The number of residues that have more than average conservation grade was calculated as follows: ~60.4% for SV2A (448/742), ~62.4% for SV2B (426/683), ~61.6% for SV2C (448/627), ~47.6% for SYP (149/313), ~54.5% for SYNGR1 (127/233), ~55.3% for SYNGR2 (124/224), ~54.1% for SYNGR3 (124/229), 50% for SYNGR4 (117/234). Conservation grades were calculated for the source amino acid sequence and the corresponding Alphafold3 predicted native structure. For clarity, the N- and C-termini and large loops were deleted in the figure.

Figure S17. Enlarged panel a-h of Figure 1. The QTY variant sequences are below the native protein sequences. The QTY amino acid substitution changes are colored in red. Other color code: Yellow line-intracellular, Blue wave-transmembrane helices, Pinkish line-extracellular.



a) SV2A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b) SV2B

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MEDSYKDRTSLMKGAKDIAREVKKQTVKKVNQAVDRAQDEYTQRSYSRFQDEEDDDDYYPAGETYNGEANDDEGSSEATE

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GHDEDDEIYEGEYQGIPSMNQAKDSIVSVGQPKGDEYKDRRELESERRADEEELAQQYELIIQECGHGRFQWAQYYTQGM

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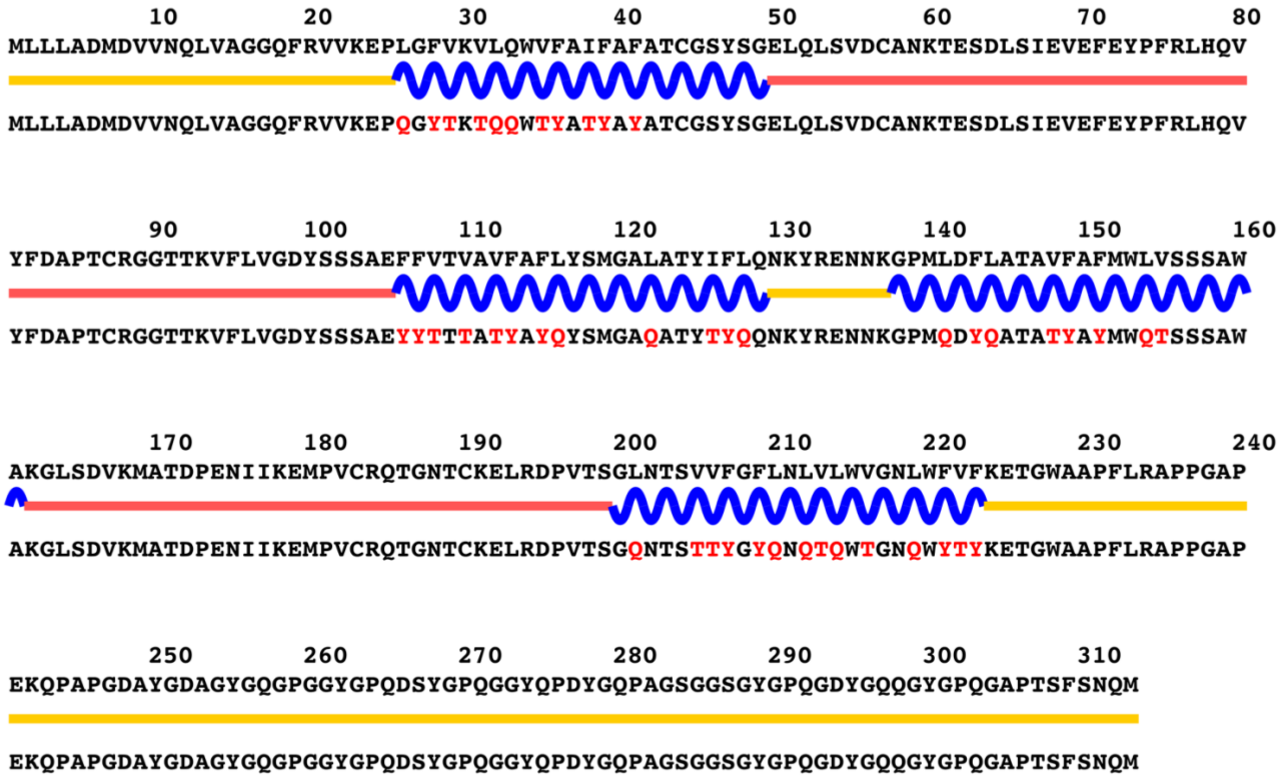
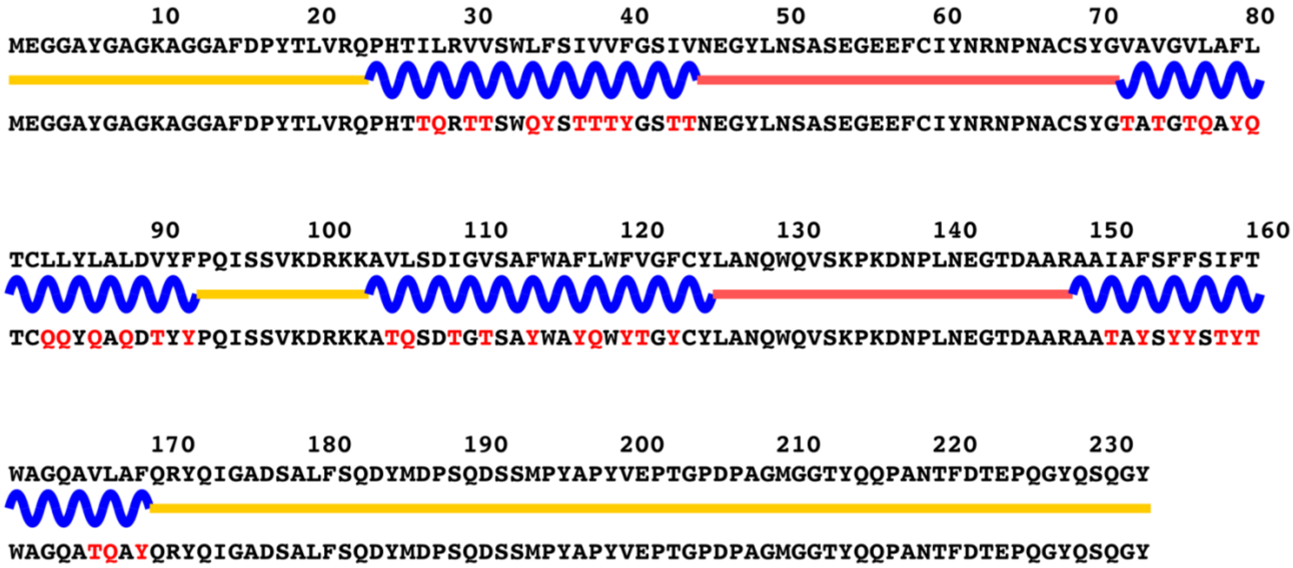
TRTQVLM

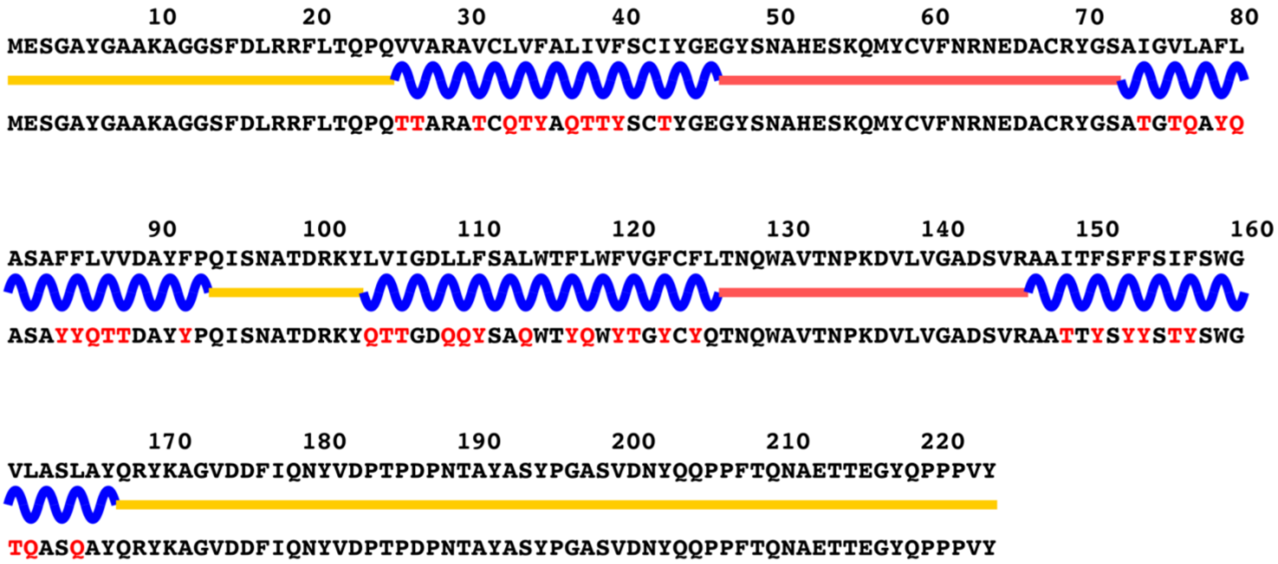
c) SV2C



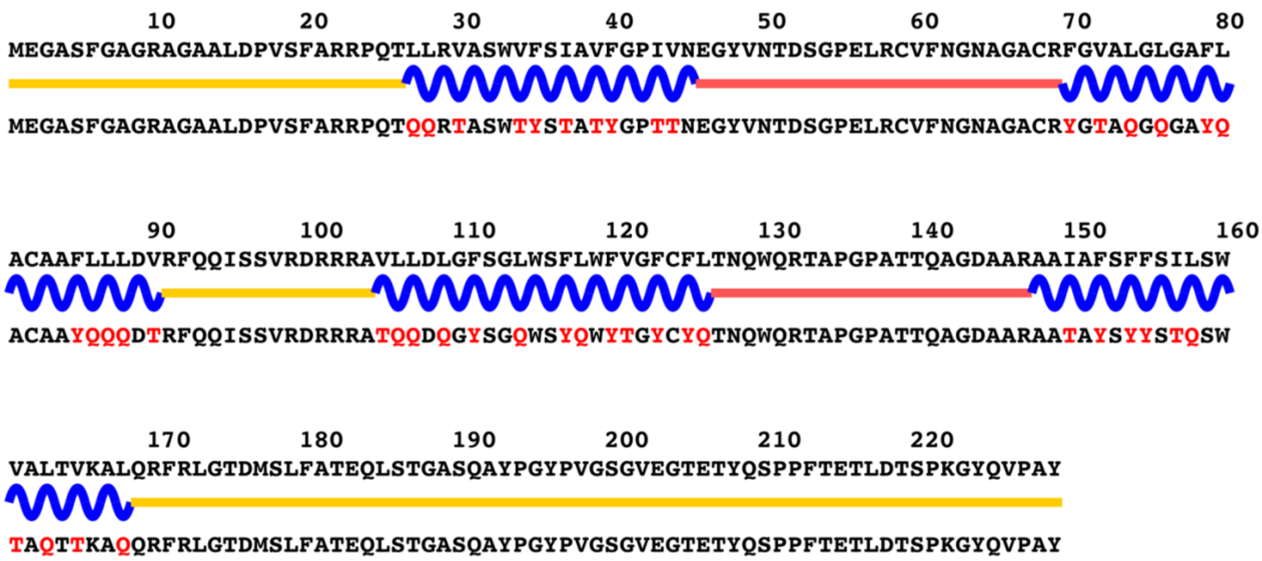
d) SYP



e) SYNGR1



f) SYNGR2



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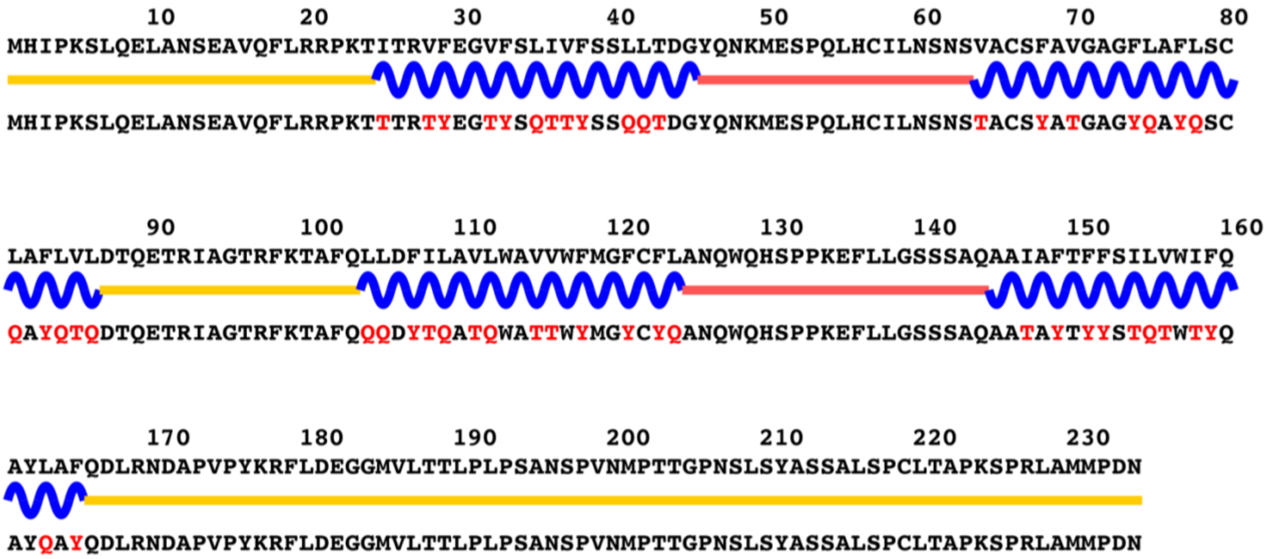
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h) SYNGR4

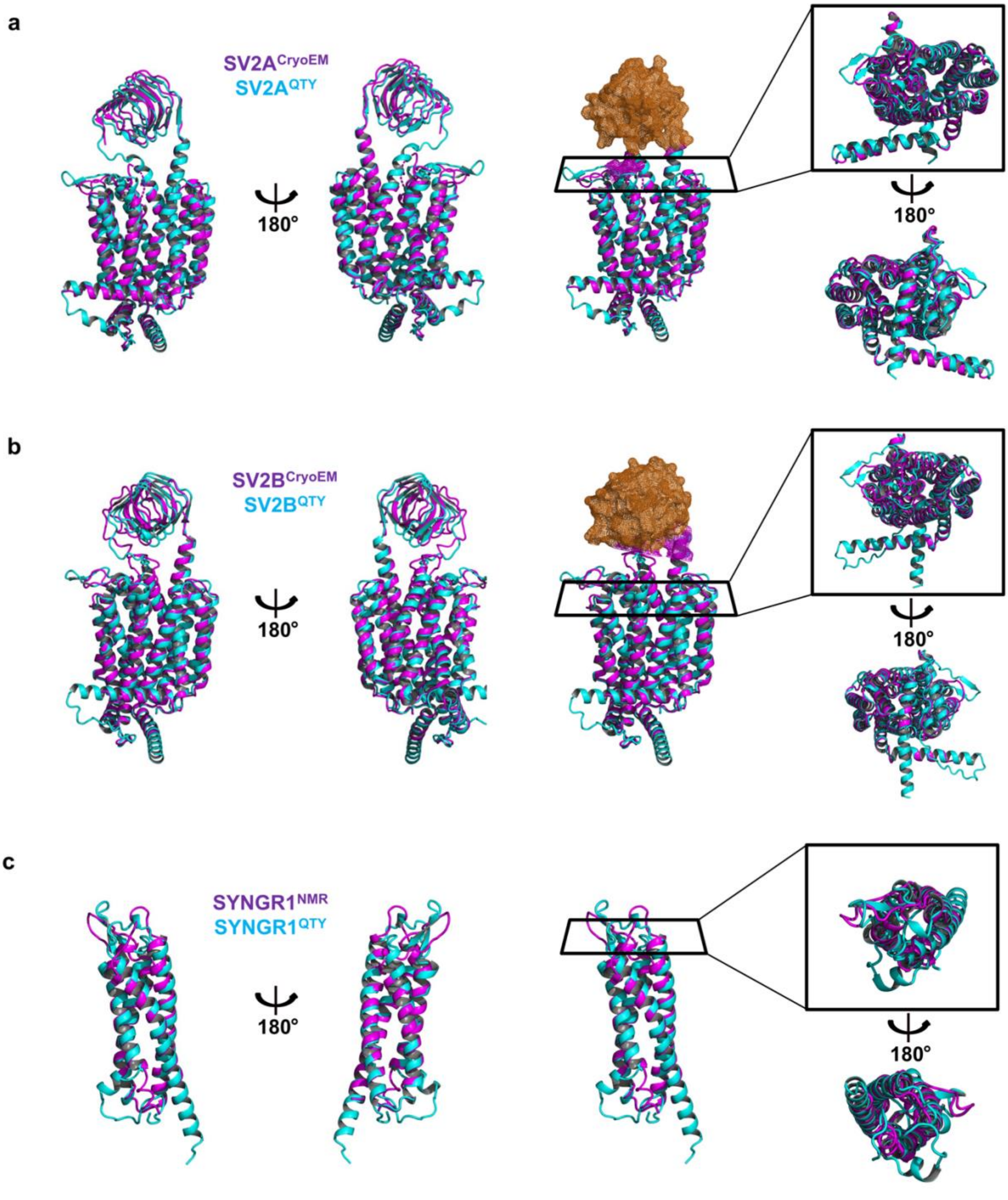

Figure S18. Superposition of experimental structures SV2A^{CryoEM}, SV2B^{CryoEM}, SYNGR1^{NMR} and their corresponding AlphaFold3 predicted water-soluble QTY variants. The experimentally determined native structures SV2A (PDB ID: 8UO9), SV2B (PDB ID: 8UO8), and SYNGR1 (PDB ID: 8A6M, state 3) were sourced from the RCSB PDB database. These native structures, shown in magenta, were superposed with their AlphaFold3 predicted water-soluble variants, depicted in cyan. The root mean square deviations (RMSD) are: **a** SV2A^{CryoEM} *vs* SV2A^{QTY} (1.313Å), **b** SV2B^{CryoEM} *vs* SV2B^{QTY} (2.748Å), and **c** SYNGR1^{NMR} *vs* SYNGR1^{QTY} (1.990Å). Unresolved large terminal loops in the experimental structures were removed to provide a clearer view for direct comparisons. The superposed structures shown from four different angles: front, back, top, and bottom. In the top and bottom views, the extramembrane structures (brown color) are hidden to highlight the comparison of the intramembrane domains.

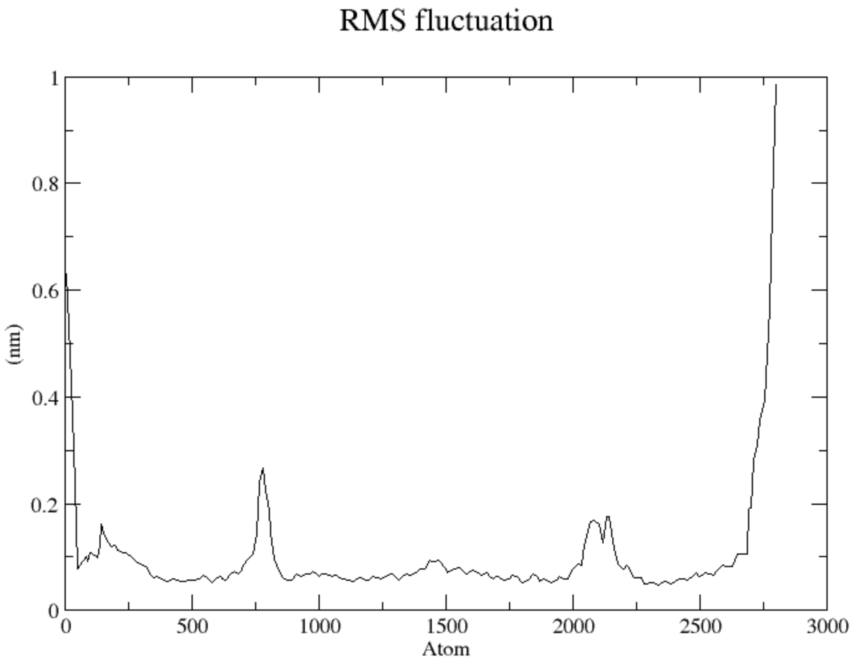

Figure S19. RMS fluctuation of SYNGR1 after 100ns MD simulation.

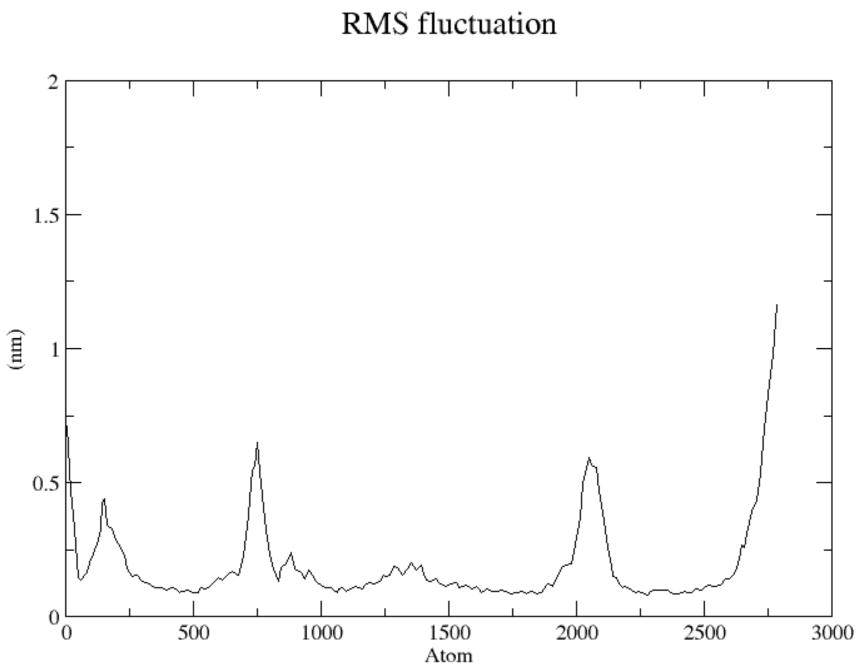

Figure S20. RMS fluctuation of SYNGR1^{QTY} after 100ns MD simulation.

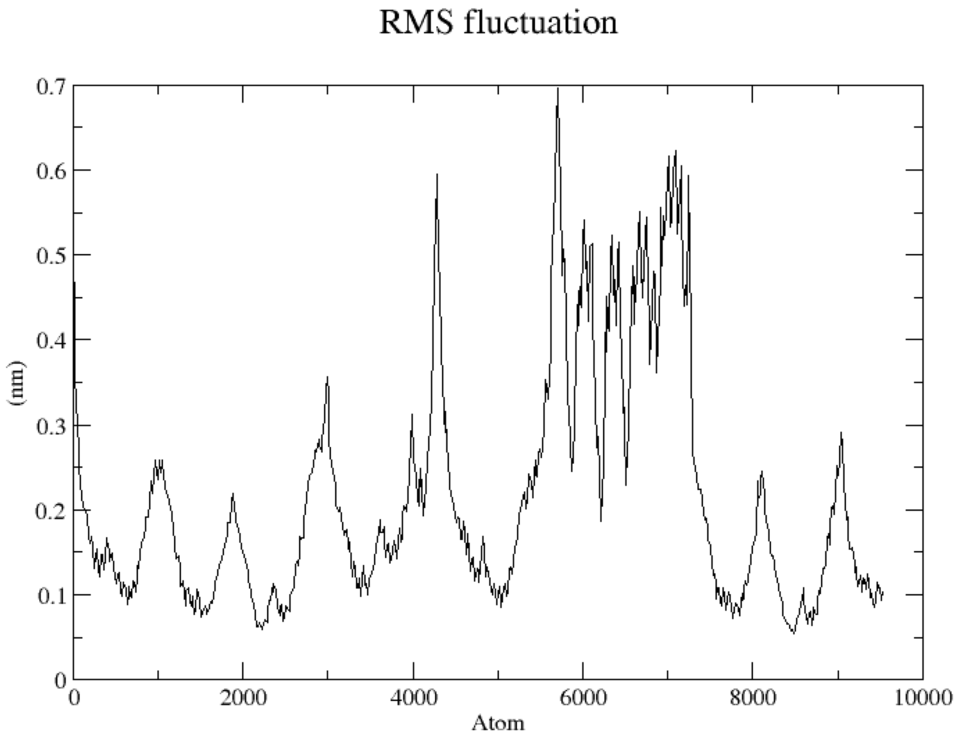

Figure S21. RMS fluctuation of SV2A after 100ns MD simulation.

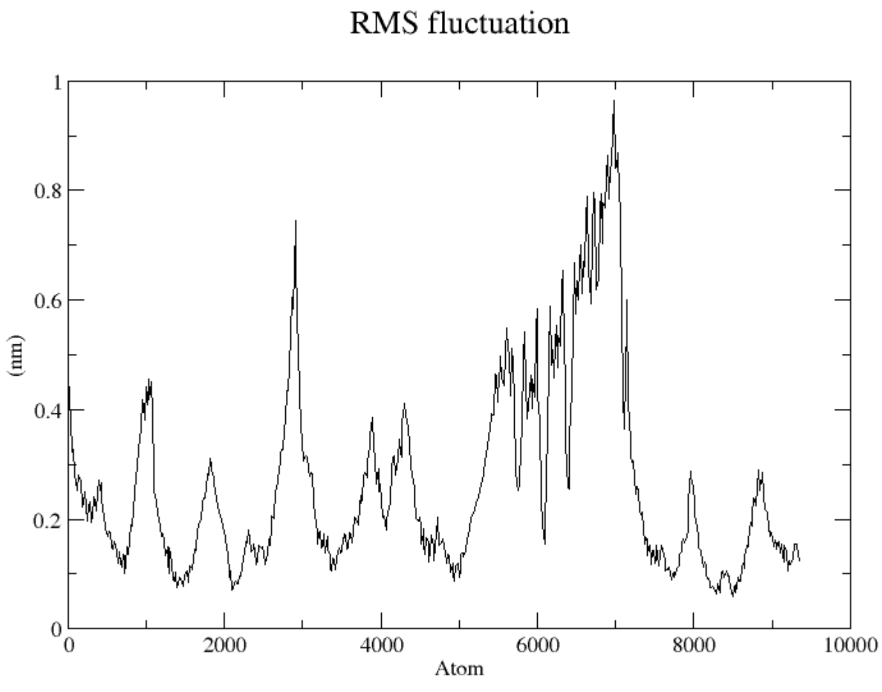

Figure S22. RMS fluctuation of SV2A^{QTY} after 100ns MD simulation.

Table S1. Natural variants of L>Q, I>T, F>Y in the 8 synaptic vesicle proteins (Each change affects the second position of the codons, V>T variants are not present).

Name	Variant[1]	2nd base[2]	Location[3]	Structure[4]	Exposure[5]	Conservation Grade[6]	Predicted Effect[7]	Clinical Significance[8]
SV2A	L148Q	U>A	Intracellular	α-helix	Buried	8	Likely pathogenic	-
	L213Q	U>A	TM2	α-helix	Buried	5	Likely pathogenic	-
	L519Q	U>A	ECL4	Loop	Buried	4	Likely pathogenic	-
	I12T	U>C	Intracellular	α-helix	Buried	4	Likely pathogenic	-
	I18T	U>C	Intracellular	α-helix	Buried	6	Likely pathogenic	-
	I93T	U>C	Intracellular	Loop	Buried	7	Likely pathogenic	-
	I101T	U>C	Intracellular	Loop	Buried	8	Likely pathogenic	-
	I269T	U>C	TM4	α-helix	Buried (S[9])	9	Likely pathogenic	-
	I302T	U>C	TM5	α-helix	Buried	5	Likely benign	-
	I348T	U>C	TM6	α-helix	Buried	7	Ambiguous	-
	I407T	U>C	ICL3	α-helix	Buried	5	Ambiguous	-
	I472T	U>C	ECL4	α-helix	Buried (S[9])	9	Likely pathogenic	-
	I517T	U>C	ECL4	β-bridge	Buried	3	Likely benign	-
	I613T	U>C	TM8	α-helix	Buried (S[9])	9	Likely pathogenic	-
	I622T	U>C	ICL4	α-helix	Buried	6	Ambiguous	-
	I703T	U>C	TM11	α-helix	Buried (S[9])	9	Likely pathogenic	-
	F247Y	U>A	TM3	α-helix	Buried	6	Likely pathogenic	-
	F435Y	U>A	ICL3	α-helix	Buried	3	Likely pathogenic	-
	F487Y	U>A	ECL4	β-bridge	Exposed	5	Ambiguous	-
	F686Y	U>A	TM11	α-helix	Buried	8	Likely pathogenic	-
SV2B	L119Q	U>A	TM1	α-helix	Buried (S[9])	9	Likely pathogenic	-
	L239Q	U>A	TM5	α-helix	Buried	8	Likely pathogenic	-
	I153T	U>C	TM2	α-helix	Buried	5	Likely benign	-
	I212T	U>C	TM4	α-helix	Buried (S[9])	9	Likely benign	Uncertain significance
	I218T	U>C	TM4	α-helix	Buried	8	Likely benign	-
	I316T	U>C	ICL3	α-helix	Buried	7	Likely pathogenic	-
	I350T	U>C	ICL3	Loop	Buried	6	Ambiguous	-
	I352T	U>C	ICL3	Loop	Buried	5	Likely pathogenic	-
	I498T	U>C	ECL4	β-bridge	Buried	1	Likely benign	-
	I515T	U>C	ECL4	β-bridge	Exposed	3	Likely benign	-

	I563T	U>C	ICL4	α-helix	Buried	6	Likely benign	-
	I575T	U>C	TM9	α-helix	Buried	5	Likely benign	-
	F203Y	U>A	TM3	α-helix	Buried	4	Ambiguous	-
	F349Y	U>A	ICL3	Loop	Buried	5	Ambiguous	-
	F581Y	U>A	TM9	α-helix	Buried	7	Ambiguous	-
SV2C	L193Q	U>A	TM2	α-helix	Buried (S^9)	9	Likely pathogenic	-
	L222Q	U>A	ICL1	α-helix	Buried	4	Ambiguous	-
	L392Q	U>A	ICL3	Loop	Buried	5	Likely pathogenic	-
	L603Q	U>A	ICL4	α-helix	Buried	7	Likely pathogenic	-
	L716Q	U>A	TM12	α-helix	Buried	7	Ambiguous	-
	I88T	U>C	Intracellular	Loop	Buried	7	Likely pathogenic	-
	I106T	U>C	Intracellular	Loop	Exposed	1	Likely benign	-
	I196T	U>C	TM2	α-helix	Buried	6	Likely benign	-
	I259T	U>C	TM4	α-helix	Buried	7	Ambiguous	-
	I291T	U>C	TM5	α-helix	Buried	7	Likely pathogenic	-
	I389T	U>C	ICL3	Loop	Exposed	2	Likely benign	-
	I393T	U>C	ICL3	Loop	Buried	5	Likely benign	-
	I395T	U>C	ICL3	Loop	Buried	4	Likely pathogenic	-
	I434T	U>C	ICL3	α-helix	Buried	1	Likely benign	-
	I483T	U>C	ECL4	β-bridge	Buried	7	Likely benign	-
	I491T	U>C	ECL4	β-bridge	Buried	5	Likely benign	-
	I503T	U>C	ECL4	β-bridge	Buried	3	Likely benign	Uncertain significance
	I553T	U>C	ECL4	β-bridge	Buried	4	Likely benign	-
	I598T	U>C	TM8	α-helix	Buried (S^9)	9	Likely pathogenic	-
	I637T	U>C	TM10	α-helix	Buried	8	Ambiguous	-
	I701T	U>C	TM12	α-helix	Buried (S^9)	9	Likely pathogenic	-
	F512Y	U>A	ECL4	β-bridge	Buried	8	Ambiguous	-
SYP	L2Q	U>A	Intracellular	α-helix	Exposed	6	Likely benign	Uncertain significance / Likely benign
	L233Q	U>A	TM4	α-helix	Exposed	2	Likely benign	-
SYNGR1	L20Q	U>A	Intracellular	α-helix	Buried	7	Likely pathogenic	-
	L106Q	U>A	TM3	α-helix	Buried	4	Likely pathogenic	-
	I27T	U>C	TM1	α-helix	Buried	6	Ambiguous	-

	I37T	U>C	TM1	α-helix	Buried	7	Likely pathogenic	-
	I109T	U>C	TM3	α-helix	Buried	6	Likely benign	-
SYNGR2	L17Q	U>A	Intracellular	α-helix	Buried	4	Likely benign	-
	L165Q	U>A	TM4	α-helix	Buried	4	Likely benign	-
	I43T	U>C	TM1	α-helix	Buried (S[9])	9	Likely pathogenic	-
	I74T	U>C	TM2	α-helix	Buried	7	Likely benign	-
	I95T	U>C	ICL1	Loop	Buried	7	Likely pathogenic	-
	F84Y	U>A	TM2	α-helix	Buried	4	Likely benign	-
SYNGR3	L217Q	U>A	Intracellular	Loop	Exposed	4	Likely benign	-
SYNGR4	I3T	U>C	Intracellular	Loop	Exposed	3	Likely benign	-
	I25T	U>C	TM1	α-helix	Buried	5	Likely benign	-
	I93T	U>C	ICL1	Loop	Buried	8	Likely benign	-
	I154T	U>C	TM4	α-helix	Buried	8	Likely benign	-
	F107Y	U>A	TM3	α-helix	Buried (S[9])	9	Likely benign	-
	F151Y	U>A	TM4	α-helix	Buried (S[9])	9	Likely benign	-
	F159Y	U>A	TM4	α-helix	Buried	1	Likely benign	-

[1]Protein consequence of the variant according to HGVS numbering.
[2]The second base of the residue codon for the corresponding variant.
[3]Topological localizations of the variants according to molecular architecture (TM = Transmembrane, ECL = Extracellular loop, ICL = Intracellular loop). The topological information of the mature protein obtained from Uniprot.
[4]Secondary structure of the corresponding residue, calculated according to Define Secondary Structure of Proteins (DSSP) algorithm from the determined models of native proteins available in the AlphaFold Database.
[5]Residue exposure according to the NACSES algorithm, predicted by ConSurf server,
[6]Evolutionary conservation grade of the residue predicted by ConSurf server; 1 to 9, in order of increasing conservation (1= Variable, 5= Average, 9= Conserved).
[7]Variant effect predicted by AlphaMissense.
[8]Based on ClinVar's January 29, 2024 release.
[9]A structural residue (buried and highly conserved) predicted by ConSurf Server.

Table S2. Natural variants of Q>L, T>I, Y>F in the 8 synaptic vesicle proteins (Each change affects the second position of the codons, V>T variants are not present).

Name	Variant[1]	2nd base[2]	Location[3]	Structure[4]	Exposure[5]	Conservation Grade[6]	Predicted Effect[7]	Clinical Significance[8]
SV2A	Q376L	A>U	ICL3	α-helix	Exposed	4	Likely benign	-
	T257I	C>U	ECL2	α-helix	Buried	4	Likely benign	-
	T352I	C>U	TM6	H-bonded turn	Exposed	4	Likely benign	Uncertain significance
	T380I	C>U	ICL3	α-helix	Exposed (F[10])	9	Likely benign	-
	T395I	C>U	ICL3	Loop	Exposed	4	Likely pathogenic	-
	T496I	C>U	ECL4	β-bridge	Buried	5	Ambiguous	-
	T540I	C>U	ECL4	β-bridge	Exposed	5	Likely benign	-
	T673I	C>U	TM10-ICL5	α-helix	Buried (S[9])	9	Likely pathogenic	-
	T711I	C>U	ECL6	Loop	Buried	8	Likely pathogenic	-
	Y46F	A>U	Intracellular	α-helix	Exposed	7	Likely benign	-
	Y212F	A>U	TM2	α-helix	Buried	7	Likely benign	-
	Y255F	A>U	TM3-ECL2	α-helix	Exposed (F[10])	8	Likely benign	-
SV2B	Q345L	A>U	ICL3	Loop	Exposed	5	Likely benign	-
	T97I	C>U	Intracellular	α-helix	Exposed	4	Likely benign	-
	T269I	C>U	ECL3	Bend	Exposed	8	Likely pathogenic	-
	T323I	C>U	ICL3	α-helix	Exposed (F[10])	9	Likely benign	-
	T330I	C>U	ICL3	Loop	Exposed	4	Likely benign	-
	T356I	C>U	ICL3	Bend	Exposed	5	Likely pathogenic	Uncertain significance
	T358I	C>U	ICL3	Loop	Exposed	5	Ambiguous	-
	T460I	C>U	ECL4	β-bridge	Exposed	3	Likely benign	-
	T487I	C>U	ECL4	Loop	Exposed	7	Likely pathogenic	-
	T497I	C>U	ECL4	Loop	Buried	4	Likely pathogenic	-
	T614I	C>U	TM10-ICL5	α-helix	Buried (S[9])	9	Likely pathogenic	-
	T625I	C>U	ICL5	α-helix	Exposed (F[10])	9	Likely pathogenic	-
	T643I	C>U	TM12	α-helix	Buried	5	Likely benign	-
	Y10F	A>U	Intracellular	Loop	Exposed	5	Likely benign	-
	Y46F	A>U	Intracellular	Loop	Exposed (F[10])	9	Ambiguous	-
	Y404F	A>U	TM7	α-helix	Buried	6	Likely benign	-

SV2C	Q219L	A>U	ICL1	α-helix	Exposed	5	Likely benign	-
	Q239L	A>U	TM3	Loop	Exposed (F^{10})	9	Likely pathogenic	-
	T26I	C>U	Intracellular	α-helix	Exposed	4	Ambiguous	-
	T64I	C>U	Intracellular	Loop	Exposed	1	Likely benign	-
	T181I	C>U	ECL1	α-helix	Exposed	6	Ambiguous	-
	T338I	C>U	TM6	α-helix	Exposed (F^{10})	5	Ambiguous	-
	T482I	C>U	ECL4	β-bridge	Exposed	5	Likely benign	-
	T486I	C>U	ECL4	Loop	Exposed (F^{10})	8	Likely pathogenic	-
	T536I	C>U	ECL4	β-bridge	Exposed (F^{10})	8	Likely benign	-
	T540I	C>U	ECL4	Loop	Buried	5	Ambiguous	-
	T545I	C>U	ECL4	Loop	Buried (S^9)	9	Likely pathogenic	-
	T590I	C>U	TM8	α-helix	Buried (S^9)	9	Likely pathogenic	-
	T611I	C>U	TM9	α-helix	Exposed	5	Ambiguous	-
	T631I	C>U	TM9-ECL5	Loop	Exposed	7	Likely benign	-
	T658I	C>U	TM10-ICL5	α-helix	Buried (S^9)	9	Likely pathogenic	-
	T706I	C>U	TM12	α-helix	Buried	3	Likely benign	-
	Y117F	A>U	Intracellular	3/10 helix	Buried	1	Likely benign	-
	Y198F	A>U	TM2	α-helix	Buried	6	Likely benign	-
	Y643F	A>U	TM10	α-helix	Buried	8	Likely benign	-
SYP	Q79L	A>U	ECL1	α-helix	Exposed	4	Likely pathogenic	-
	Q265L	A>U	Intracellular	Loop	Exposed	7	Ambiguous	-
	T198I	C>U	ECL2	Loop	Buried	7	Likely benign	Likely benign
	T203I	C>U	TM4	α-helix	Buried	7	Likely pathogenic	-
	T307I	C>U	Intracellular	Loop	Exposed	6	Likely pathogenic	-
SYNGR1	Q129L	A>U	ECL2	α-helix	Exposed	7	Likely pathogenic	-
	T19I	C>U	Intracellular	α-helix	Exposed	4	Likely benign	-
	T26I	C>U	TM1	α-helix	Buried	8	Likely pathogenic	-
	T81I	C>U	TM2	α-helix	Buried	6	Likely benign	-
	T213I	C>U	Intracellular	Loop	Exposed	3	Likely benign	-
	Y61F	A>U	ICL1	H-bonded turn	Buried	6	Likely benign	-
	Y70F	A>U	ICL1	α-helix	Buried	8	Likely benign	-

SYNGR2	Q25L	A>U	Intracellular	α-helix	Exposed	5	Likely benign	-
	T116I	C>U	TM3	α-helix	Buried	6	Ambiguous	-
	T133I	C>U	ECL2	Bend	Exposed	7	Likely pathogenic	-
	T185I	C>U	Intracellular	Loop	Exposed	4	Likely benign	-
	T190I	C>U	Intracellular	H-bonded turn	Exposed	4	Likely benign	-
	Y181F	A>U	Intracellular	α-helix	Exposed	5	Likely benign	-
SYNGR3	T26I	C>U	Intracellular	α-helix	Exposed	5	Likely pathogenic	-
	T182I	C>U	Intracellular	α-helix	Exposed	5	Likely pathogenic	-
	T207I	C>U	Intracellular	Loop	Exposed	4	Ambiguous	-
	T214I	C>U	Intracellular	Loop	Exposed	5	Ambiguous	-
	T216I	C>U	Intracellular	Loop	Exposed	4	Ambiguous	-
	T219I	C>U	Intracellular	Loop	Exposed	3	Likely benign	-
SYNGR4	Q89L	A>U	ICL1	α-helix	Buried	5	Likely benign	-
	T24I	C>U	Intracellular	α-helix	Exposed (F[10])	8	Likely benign	-
	T43I	C>U	TM1	α-helix	Exposed (F[10])	8	Ambiguous	-
	T96I	C>U	ICL1	α-helix	Buried	6	Likely benign	Uncertain significance
	T150I	C>U	TM4	α-helix	Buried (S[9])	9	Likely benign	-
	T189I	C>U	Intracellular	Bend	Buried	5	Likely benign	-

[1]Protein consequence of the variant according to HGVS numbering.
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[7]Variant effect predicted by AlphaMissense.
[8]Based on ClinVar's January 29, 2024 release.
[9]A structural residue (buried and highly conserved) predicted by ConSurf Server.
[10]A functional residue (exposed and highly conserved) predicted by ConSurf Server.

Table S3. QTY-evolutionary profiling results, correlations between conservation grades and RMSD values (100ns equilibrated simulation, between QTY-variants and native proteins).

Name	All residues	Only Buried Residues	Only Exposed Residues
SYNGR1	rho(144)= –0.345, $p < .001$	rho(79)= -0.102, $p = .364$	rho(63)= -0.46, $p < .001$
S2VA	rho(607)= –0.252, $p < .001$	rho(374)= -0.045, $p = .384$	rho(231)= -0.376, $p < .001$